# Full Instructor Guide

EM Evidence Sim · updated 10 October 2026

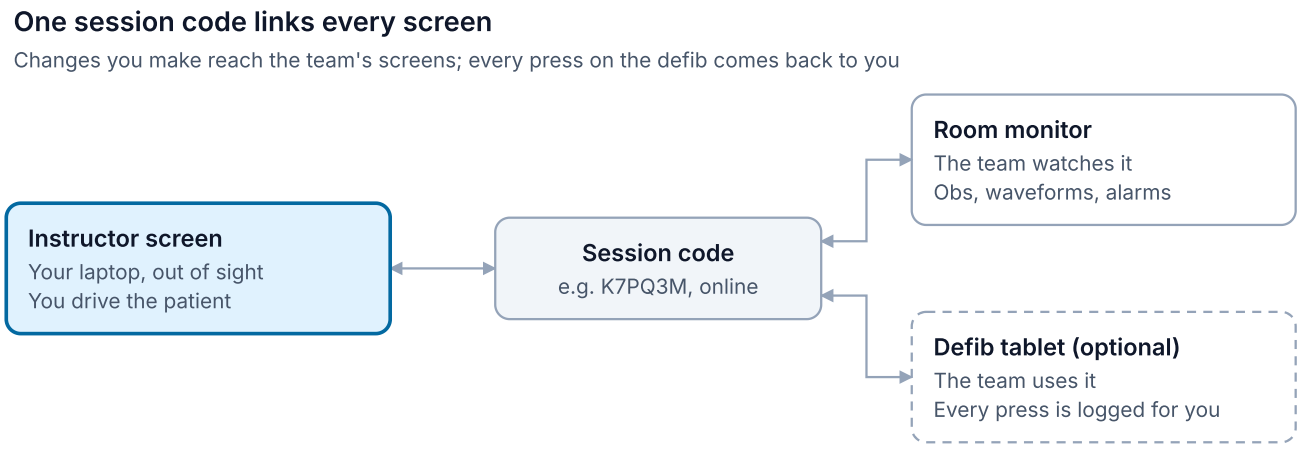
This guide explains everything an instructor needs to set up, run and debrief a scenario. New to the app? Start with the Quick Start Guide.

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## At a glance

A simulation uses up to three screens, all linked online by one six-character session code.




How the screens link: three devices, one session code.

You drive the patient from the instructor screen. The room monitor shows the team only what they have attached. The defib tablet, if used, sends every press back to your event log.

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## Before the session

Allow 10 minutes to set up the screens and test them before the team arrives.

### Equipment

| Device | Role | Where it goes |
|---|---|---|
| Laptop or tablet | Instructor screen: you drive the patient from here | Out of the team's sight, ideally where you can see the room |
| Second screen, TV, projector or device | Room monitor: the patient monitor the team watches | Where a real monitor would be, visible to the whole team |
| Tablet (optional) | Defibrillator | At the bedside, for Defib Sim or any scenario that needs a defib |
| Tablet (optional) | Ventilator (HAMILTON-T1) | At the bedside, for Ventilator Sim or any ventilated patient. It can also show on the room monitor tablet |
| Speakers (optional) | Alarms and monitor beeps | Next to the room monitor |

All devices need an internet connection, and any up-to-date web browser. You do not need an account to run the built-in scenarios.

### Pre-flight checklist

- ☐ Open emsimulator.com on the instructor laptop and note the session code
- ☐ Open the room monitor and tap **Tap to Enable Sound**
- ☐ Link the defib tablet, if you are using one
- ☐ Check the badge on the instructor screen reads **Live** and names the linked screens
- ☐ Load your scenario and read its brief
- ☐ Turn the room monitor's volume up and check you can hear an alarm
- ☐ Change one vital sign and check the room monitor follows

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## Session codes and linking screens

The six-character session code (for example K7PQ3M) links every screen in one simulation. It never uses look-alike characters (no 0/O or 1/I/L), so it is easy to read out. It shows at the bottom of the start screen and as "Session ID" in the top right corner during a scenario.

- **Same code all day:** one code can run any number of scenarios with the same group. The linked screens stay linked between scenarios.
- **New code:** open **Connect patient screens** on the start screen and press **New code** for a new group. Screens on the old code must join again.

### Ways to link a screen

| Screen | On the same laptop | On a separate device |
|---|---|---|
| Patient screens | On the start screen press **Connect patient screens**, choose how many patient screens (1, 2 or 3) and **Plugged into this computer**, then press **Open screen 1**, **Open screen 2** and so on (or, once running, the same items under **Screens**). Each opens in its own window: drag it onto its screen and make it full screen. Keep them as windows, not tabs | Choose the number of screens and **Separate devices**, then scan each screen's QR code with that device; or, on the device, open the site, choose **Patient screen** when it asks what the device is for, type the code, pick what it shows and press **Connect** |
| What each screen shows | **1 screen:** the monitor, switching to the defib or ventilator when you bring them in. **2 screens:** the monitor, and a screen with the defib and the ventilator. **3 screens:** the monitor, the defib and the ventilator. Your **Defib** and **Ventilator** buttons bring a device in; it comes to the front on the screen that shows it, and the candidate can switch with the buttons at its bottom left. A device stays running while it is switched away from | |

**Checking the link:** the badge in the instructor top bar shows the connection.

| Badge | Meaning |
|---|---|
| Live · no screens linked | Your session is online but nothing has joined yet |
| Live · patient monitor (or "2 screens linked") | Linked and working |
| Syncing | Connecting; wait a few seconds |
| Sync error | The connection has failed; check the internet |

Hover over the badge to see each linked screen and when it was last heard from.

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## Choosing a scenario

When you open the app it asks **What is this device for?** Choose **Controller** on your laptop (a patient screen chooses **Patient screen**). The start screen then asks **What do you want to run?** with a card for each way to start; **All options** takes you back to the cards. The Scenario library covers most teaching; Defib Sim is for defibrillator skills and Ventilator Sim for ventilator skills. Setting up the patient screens is optional before you start: with nothing set up the app uses one screen, and **Connect patient screens** (or **Screens** during a scenario) sets them up whenever you are ready.

| Card | What you get | Best for |
|---|---|---|
| **Scenario library** | 254 written scenarios in 8 categories: Adult Medical, Trauma, Paediatric, Cardiac Arrest, Toxicology, Obs & Gynae, Elderly, Psychiatric | Planned teaching with set objectives |
| **Quick Sim** | A blank patient: you set age, weight, sex, name and starting rhythm. No scenario, drugs or scoring | Ad-hoc teaching, rhythm recognition, obs drills |
| **Defib Sim** | Defibrillation, cardioversion and pacing on the defib tablet, with its own controller | Defib skills and assessment (see Running a Defib Sim) |
| **Ventilator Sim** | Nine ventilator scenarios on the HAMILTON-T1: NIV, CPAP, lung-protective and asthma ventilation, setting up after RSI, troubleshooting, mucus plugging and transfer, with the full instructor screen | Ventilator skills and assessment (see The ventilator and Ventilator Sim) |
| **Random scenario** | A generated patient from your choice of category, age group and acuity | Unpredictable cases for experienced teams |
| **My scenarios: Saved** | Scenarios you have saved or imported as a file. **Edit**, **Load** or **Delete** each one (Delete asks you to confirm and removes it from that device) | Your own material |
| **My scenarios: Builder** | A form to write a scenario field by field | Writing new scenarios |
| **Restricted** (in the Scenario library) | Copyright-restricted scenarios (for example RCUK) | Only if your account has been given access |

**Human factors.** Random and Builder can add a challenge: Blindfolded Lead, Silent Team, New Junior, Missing Kit or Distracted Senior. It shows as a badge on your screen so you can brief the right person.

**Paediatric scenarios.** WETFLAG starts hidden on the room monitor. Tick **Show WETFLAG on Monitor from the start** on the start screen (under the scenario options) if you want the team to see it straight away. You can show or hide it at any time during the scenario (see Paediatric scenarios).

**The brief.** Loading a library, random or saved scenario opens a brief before anything starts:

- **Patient Brief:** name, history, medicines and allergies
- **Required Equipment**
- **Clinical Progression:** how the scenario is expected to unfold
- **Key Interventions** the team should perform
- **Guidelines & Resources**
- **Learning Objectives:** what the debrief scores against

Press **Start Scenario** to begin, or **Back to Setup** to choose another.

**Resume Previous?** If the instructor tab was closed during a scenario, the setup screen offers to resume it. The app saves your progress every few seconds.

**Scenario paused.** Pressing **Back** during a scenario pauses it and returns to the setup screen, which shows **Return to scenario**. Press it to carry on where you left off, then **START**. Starting a different scenario replaces the paused one.

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## A tour of the instructor screen

The screen has a top bar, the patient on the left, and treatments and the event log on the right. Drag the divider between the columns to widen either side.

### Top bar

| Control | What it does |
|---|---|
| **Back** | Pauses the scenario and returns to the setup screen; **Return to scenario** there brings you back |
| **Finish** | Ends the scenario and opens the debrief (asks you to confirm) |
| **PAUSE / START** | Stops or restarts the clock |
| Connection badge | Shows the session is live and which screens are linked |
| **Safety flags** | Appears when the team does something out of the usual order; opens the list |
| **Muted** | Appears in red while alarms are muted; press to unmute |
| **Simple / Full** | Hides or shows the advanced controls (below) |
| **Screens ▾** | **Join by QR code** (a QR code for each kind of screen: monitor, defib, ventilator), and **Open screen 1: …**, **Open screen 2: …** for each patient screen in your set-up, in a new window on this computer |
| **Tools ▾** | Drug calculator, Timer alerts, Full simulation log, where sound plays, Mute alarms, Keyboard shortcuts |

### Left column: the patient

Each grey bar is a section. Click its title to open or close it; a closed section shows a one-line summary. The app remembers which sections you keep open on that device.

| Section | Starts | Contains |
|---|---|---|
| Patient details | Always shown | Name, age, scenario, diagnosis, story, any human factors challenge |

| Section | Starts | Contains |
|---|---|---|
| Monitoring & access | Closed | What is attached (ECG, sats, cuff, temperature, capnography, arterial line, IV access); **Attach** stays on the bar |
| Monitor strip and vitals | Always shown | The waveforms, **Show timer**, **Change rhythm**, and tiles for HR, BP, SpO2, RR, Temp and ETCO2 |
| More obs | Open | GCS and glucose (plus pH and K+ in Full view) |
| Drugs on board | Open, once a drug is given | Each drug's phase, strength and time left |
| Patient condition | Closed | AUTO or MANUAL deterioration; **Better** and **Worse** stay on the bar |
| Rhythm & resus | Open | ARREST, ROSC, Arrest View, Defib, Cycle NIBP Now, Auto NIBP, Manual Reading, Show/Hide WETFLAG on Monitor (child scenarios), and the rhythm and shock tally |

## Right column: treatments and the log

- **Search** finds any treatment by name or route (for example "IM").
- **Tabs:** Common, Drugs, Airway, Breathing, Circulation, Procedures, Investigations, Voice and Assessment. In Common, the scenario's **Recommended actions** sit at the top in an amber box.
- **Voice:** type what the patient says and press **Speak**; it is read aloud on whichever screen plays sound.
- **Assessment:** tick or cross Safe Approach, Team Leadership, Communication, CPR Quality, Defib Safety and Re-evaluation as you watch.
- **Event log** (bottom): always on screen; see The event log, notes and flags.

## Simple and Full view

Simple is the default and is remembered on each device. It hides:

- the drug-timing detail lines in Drugs on board
- the "+ Invasive" button that attaches IV access, an arterial line and capnography in one go (the separate buttons stay)
- the pH and K+ tiles
- the defib's shock-response settings and "Next shock converts to" (one line shows what is set; click it to switch to Full)
- the Quick Sim preset list (a preset that is running stays on screen)

Nothing is removed in either view.

**On a phone,** the obs tiles, rhythm, clock and START/PAUSE sit at the top, and the page scrolls.

## Running the patient

You mirror what the team does: they act in the room, you press the matching control, and the patient and room monitor respond.

### Monitoring and access

The room monitor only shows what the team has attached. Open Monitoring & access to see a chip for each item.

- Click a chip to attach it (○) or remove it (●). Removing one blanks its trace and number on the room monitor. Every change is logged.
- **Attach** puts on the standard four: ECG, SpO2 probe, NIBP cuff and temperature probe. Capnography, the arterial line and IV access are separate, deliberate steps.
- **Remove all** takes every monitor off. IV access stays, because it is a route, not a monitor.
- **POC glucose** and **POC VBG** take a one-off reading at that moment. Click again to take a new one.

### Vital signs

Your tiles always show the patient's true values, even when the team cannot see them; "not on monitor" means the team cannot see that number yet.

1. Click a vital's tile.
2. Type the new value. For ETCO2 you can also pick the capnogram shape: Auto, Obstructive, Force normal, Rebreathing or Curare cleft (see Capnography).
3. Under **Get there**, choose **Now**, **30 s**, **2 min** or **5 min** for a gradual change. BP starts on **Now**; everything else starts on 30 s.
4. Press **Confirm**. For BP there are two buttons instead (see below).

A heart rate you set yourself is kept if you later change the rhythm. ARREST and ROSC reset it.

### Blood pressure

Changing the BP never sends it to the team by itself. The room monitor and the defib tablet keep showing the last cuff reading until the cuff cycles, as in real life. The cuff cycles when the team presses **Cycle** on the monitor or **NIBP** on the defib, when you press **Cycle NIBP Now**, or every 3 minutes while auto is on.

When you set a BP with **Now** and the cuff is on, you choose:

- **Change, don't send**: the patient's BP changes now. The team sees it the next time the cuff cycles.
- **Change and cycle cuff now**: the BP changes and the cuff starts measuring straight away. The cuff noise plays and the new reading appears after about 5 seconds.

If you pick 30 s, 2 min or 5 min instead, the BP drifts to the new value and the team sees wherever it has got to at the next cuff cycle.

**Auto NIBP: on/off**, next to Cycle NIBP Now, is the same switch as **Auto** on the room monitor: the cuff measures every 3 minutes. It also counts down in Quick Sim, where you never press START. An arterial line, once attached, shows the BP continuously. **Manual Reading** shows a one-off cuff reading on the room monitor without changing the patient (for example a spurious reading).

## Capnography

Once capnography is attached, the room monitor draws the capnogram at half the speed of the ECG, as the ZOLL R Series does. Several breaths fit on screen, and each one is wide enough to read its shape. The height of the plateau always matches the ETCO2 number.

| Shape | When you see it |
|---|---|
| Normal: a sharp rise, a flat plateau ending at the ETCO2, then a sharp drop to zero as the patient breathes in | Normal breathing or ventilation. A slow rate gives wide waves with a long plateau; a fast rate gives narrow ones |
| Obstructive ("shark fin"): a slow, sloping rise with no flat plateau | Bronchospasm. On Auto it follows the scenario's bronchospasm and softens as bronchodilators work; Obstructive forces it |
| Rebreathing: the trace does not return to zero between breaths | Choose Rebreathing |
| Curare cleft: a notch in the plateau | Choose Curare cleft |
| No waveform: a flat line at zero | Shows by itself when nothing is moving gas: apnoea, an oesophageal tube or a disconnected circuit |
| Low trace with small ripples from chest compressions, a breath every 6 seconds | During CPR (10 ventilations a minute) |

The ETCO2 control (tap the ETCO2 tile) chooses the shape: Auto, Obstructive, Force normal, Rebreathing or Curare cleft.

## Treatments and drugs

Find the treatment by search or tab and click it. The app then:

- logs it with the time
- applies its effect over a realistic time course: onset, peak and wearing off
- shows it in **Drugs on board** with its phase (for example ONSET), strength and time left
- for infusions and oxygen, keeps it running until you click it again; use **+** and **–** in Drugs on board to change an infusion's rate

If the team gives a drug before its prerequisite (for example an IV drug with no IV access), the app records a **safety flag**. Child doses follow the RCUK paediatric guidance (see Paediatric scenarios).

### Investigations

The team can tap **12-Lead**, **VBG**, **CXR**, **Urine**, **POCUS** or **CT Head** on the room monitor to see results. From the **Investigations** tab you can also send a result yourself:

- **Scenario finding**: the scenario's own result. The box under the button shows what it says (and its image) before you send it.
- **Results with a real image**: real X-rays, CT scans, 12-lead ECGs and ultrasound images, each with a written report. Untick **Show the written report with the image** to send the image alone, so the team has to interpret it.
- Written reports without an image, or a result you type.

On the room monitor an image appears with its report; tap it to fill the screen. Many scenarios now come with their own real image (for example a tension pneumothorax, subarachnoid haemorrhage or anterior STEMI). When the team taps **12-Lead** in a scenario with a real ECG, they see that ECG while the patient is still in that rhythm; after a change of rhythm (for example into VF) the monitor draws the 12-lead from the live rhythm instead. The images are real, openly licensed clinical images from Wikimedia Commons; their authors are credited under each image and on the image credits page.

### Deterioration

**Patient condition** shows the mode:

- **AUTO:** the patient gets worse on their own, at the rate the scenario sets, unless the team treats them.
- **MANUAL:** the obs only change when you change them.

**Better** and **Worse** move the patient towards the scenario's improved or worsened obs over 30 seconds, in either mode. In many scenarios, giving one of the key treatments starts the improvement by itself, and the log says "Patient condition IMPROVING".

**Quick Sim** has no scenario, so it leaves out treatments, drugs, deterioration and scoring. Its right side holds the full rhythm list, presets (Full view), Better and Worse, and the event log.

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## Arrests, rhythms and the defibrillator

All rhythm and resuscitation controls are in the **Rhythm & resus** section.

| Control | What it does |
|---|---|
| **ARREST ▾** | Pick the arrest rhythm. The obs drop to arrest values and the arrest timer starts |
| **ROSC ▾** | Pick the post-ROSC rhythm. Circulation returns |
| **Change rhythm** (above the vitals) | Change to any rhythm without an arrest or ROSC, for example sinus to AF |

| Control | What it does |
|---|---|
| **Arrest View** | Switches the room monitor to the arrest layout: Lead II, pads and, when capnography is attached, the capnogram with the ETCO2. It does not open a defib |
| **Defib** | Opens a working defibrillator on the room monitor, with the obs still visible, and a defib panel on your screen |

The panel under these buttons shows the current rhythm (marked shockable, sync cardiovert or pulseless), the number of shocks, total and last energy, and whether CPR is running.

**Using the defib panel on your screen:** choose the energy (the recommended energy is marked), then **Charge** and **SHOCK**. **SYNC** switches synchronised cardioversion on and off. **CPR start/stop** and **Rhythm check +2m** (jump forward two minutes) are there too, with the CPR cycle timer.

**What a shock does.** By default a realistic model decides: the energy, the rhythm, whether CPR is running and the drugs given all count. In Full view you can override it:

- **Next shock converts to:** the next shock gives exactly the rhythm you pick.
- **Shock response:** converts on the 1st to 5th adequate shock, never (refractory), Auto or the realistic model.
- **Unsync shock with a pulse:** whether an unsynchronised shock into a perfusing rhythm causes VF (never, sometimes or always).
- **After ROSC:** whether VF comes back.
- **Cardioversion succeeds at:** the lowest energy at which cardioversion works.

When a shock converts the rhythm, or fails to, a pop-up tells you on your screen only. The team never sees it.

**Learner defib on a tablet.** For defib skills with a real-looking device, use the defib tablet (see Running a Defib Sim). It works in a normal scenario too: scan the **Defib** QR code from **Screens → Join by QR code** on the tablet, or, if your set-up has a defib screen, open it with **Screens → Open screen N**.

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## Running a Defib Sim

In a Defib Sim the learner works a ZOLL-style defibrillator on a tablet, and you run the patient from the **Defib controller**, which replaces the normal instructor screen.

### Setting it up

1. On the setup screen, choose the **Defib Sim** card.
2. Choose the mode:
   - **Education:** the defib shows pulse-check results and RCUK hint cards.
   - **Assessment:** no pulse-check results or hint cards, as on a real defib. Feedback comes in the debrief.

3. Choose a scenario: Ventricular Fibrillation, Pulseless VT, VT with a pulse (unstable), SVT with life-threatening features, Fast AF with life-threatening features, Complete heart block, Symptomatic bradycardia, **Free play** (starts in sinus rhythm; you change it), or **Custom scenario**.
4. Check age, weight and sex. A patient under 18 gets the paediatric hint cards, energies and drug prompts.
5. Press **Start Defib Sim**, then link the tablet with **Join** (QR code) or **Open defib**.

**Custom scenarios** run a sequence of up to five rhythms. Each step moves on at a trigger you pick: on analyse, on shock, on pacing capture, or after 30 s, 60 s or 2 min. Save, Load, Export file and Import file keep sequences on that device or share them. During the run, **Next step** and **Restart sequence** let you move it on by hand.

**The clock.** It starts when you press **START**, or by itself at the learner's first press on the defib. A banner warns you while it is not running. If you **PAUSE**, presses are still logged but the clock stays stopped until you press START.

## The Defib controller

| Area | What it does |
|---|---|
| Patient | Who the patient is, plus tags such as Needs SYNC, Needs pacing and the RCUK energy advice |
| What the pads show | The rhythm and the patient's true obs (the BP here is the live BP, not the last cuff reading) |
| The learner's defib | A live mirror of the tablet: mode, energy, SYNC, pacer settings and the screen message, plus shocks given |
| Rhythm | Click a rhythm to change it; **All rhythms** shows the full list |
| Shock response (closed) | How shocks resolve, and **Cardioversion succeeds at** (see Arrests, rhythms and the defibrillator) |
| Pacing & artefacts (closed) | The capture threshold (± 5 mA; a pulse needs about 10 mA more), and movement, mains interference or lead-off artefact |
| Arrest | Time in arrest and shocks, the cycle timer, **Start CPR**, **Metronome**, **Rhythm check (+2:00)**, **ROSC**, **Capnography** and **Cycle NIBP**. Capnography on puts a capnogram under the ECG and the ETCO2 number on the learner's defib; while it is off, the defib shows "--" for ETCO2, as a real one without its CO2 sensor does. The defib's NIBP shows only the last cuff reading: it changes when the learner presses **NIBP** on the defib or you press **Cycle NIBP** |

| Area | What it does |
|---|---|
| RCUK prompts | When adrenaline or amiodarone is due. You see these in both modes; the learner never does |
| Drugs (closed) | Adrenaline, amiodarone, atropine, adenosine, isoprenaline, adrenaline infusion, fentanyl and midazolam |
| Event log | Every press on the defib with its time; add notes and flags |

Closed sections show a one-line summary, for example "Never converts · cardioversion at 200 J".

**How shocks resolve by default.** Defib Sim uses **Auto**: an arrest converts on the third adequate shock, and a cardioversion on the first adequate synchronised shock. "Adequate" means at least 150 J for an adult (3 J/kg for a child) to defibrillate, and 70 J (1 J/kg) to cardiovert. These thresholds are simulator settings, not guideline values. An unsynchronised shock into a rhythm with a pulse causes VF.

**Pacing** captures electrically at the scenario's threshold (it varies by up to 15 mA each run) and gives a pulse about 10 mA above it. Demand pacing is inhibited by a faster native rate.

**Debrief.** The debrief lists good practice and areas for improvement: pulse checks, mode, SYNC, energies, time to first shock, drug timing, sedation before cardioversion, analgesia for pacing and capture. **Certificate** prints a certificate of completion.

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## The ventilator and Ventilator Sim

The simulator includes a HAMILTON-T1 ventilator: the same device as the standalone ventilator trainer. It works in any scenario, and Ventilator Sim builds scenarios around it. The physiology is a teaching model, not validated physiology.

### Where the candidate uses it

- **On the room monitor tablet.** In the controller's **Ventilator** section, press **Show the ventilator on the patient screens** (Ventilator Sim does this for you; the same button then reads **Take the ventilator off the patient screens**). The candidate can flip between the ventilator and the obs without stopping it. The defib and the ventilator share that screen, so opening one closes the other.
- **On its own tablet.** Scan the **Ventilator** QR code from **Screens → Join by QR code**, or, if your set-up has a ventilator screen, open it with **Screens → Open screen N**. Monitor, defib and ventilator tablets can all run at once, in sync.

### Starting a Ventilator Sim

1. On the setup screen, choose the **Ventilator Sim** card.
2. Choose **Education** (the ventilator shows its alarm troubleshooting help) or **Assessment** (no help on the screen; feedback in the debrief).

3. Choose a scenario: AECOPD (start NIV), cardiogenic pulmonary oedema (CPAP), obesity hypoventilation (NIV), ARDS (lung-protective ventilation), severe asthma (intubated), after RSI in the ED (set up from scratch), troubleshooting (DOPES), mucus plugging (bronchoscopy teaching) or interhospital transfer. Intubated scenarios start with the tube recorded and the patient sedated and paralysed.
4. Press **Start Ventilator Sim**. The ventilator opens on the room monitor; link a tablet instead if you prefer.
5. Press **START** when the candidate begins, so the event log, the obs graph and the debrief have real times.

The full instructor screen is available, with the scenario's brief at the top of the **Ventilator** section and its likely treatments in **Recommended actions**: needle decompression and finger thoracostomy, bronchoscopy, suction, sedation and paralysis, nebulisers and the rest of the treatment list. Re-running the same scenario gives the candidate a fresh ventilator.

## What the ventilator does to the patient

While it is ventilating, SpO2 follows the oxygen and PEEP, CO2 follows the ventilation and settles over 3 to 5 minutes, the RR is the ventilator's rate, and low SpO2, high CO2 and air trapping change HR and BP. This happens whether or not the clock is running, and not in a cardiac arrest. Any obs you type yourself stay as you set them until you press **Release to ventilator**; **I set them** turns the link off altogether.

## Controlling it

| Control | What it does |
|---|---|
| Lungs and Own breathing | The lung profile (normal, AECOPD, pulmonary oedema, obesity hypoventilation, ARDS, severe asthma) and whether the patient breathes for themselves |
| Lungs (adjust) | Compliance, resistance, CO2 production, oxygenation impairment and the patient's own rate |
| Problems | One press to inject, again to fix: leaks, disconnection, coughing, kinked expiratory limb, apnoea, oxygen failure, mains lost, tiring, cracked circuit, displaced tube (no CO2), blocked tube, bronchospasm, tension pneumothorax (needle, finger thoracostomy or a drain fixes it), a low battery, mucus plugging (SpO2 in the high 80s on 100% oxygen, high pressures, small breaths; the Bronchoscopy intervention or the **Bronchoscopy done** button clears it) and breath stacking (AutoPEEP builds and the BP falls). Grouped as tube and circuit, lungs, patient and equipment; once a problem is in, the panel says what the candidate should see. |
| Education / Assessment | Shows or hides the ventilator's alarm help |

| Control | What it does |
|---|---|
| Control this ventilator | Under each ventilator's mirror: switch on or off, start or standby, mode, every setting and alarm limit (step or type a value), silence, O2 enrichment, lock the screen and a fresh ventilator. Use it to hand over a ventilator with deliberate errors. Your changes are logged as yours |

**Debrief.** Any session that used the ventilator gets **Ventilator feedback**: when ventilation started, the pre-use check, alarm limits, tidal volume per kg of ideal body weight, plateau and driving pressure, AutoPEEP, SpO2 time in target, and when each problem was injected and how long it took to fix (for example "fixed 2 min 10 s later"). A Ventilator Sim debrief is headed with the scenario and its mode and is not scored against learning objectives. The targets are the simulator's teaching targets for each scenario, not a guideline: the troubleshooting scenarios (DOPES, mucus plugging, transfer) accept up to 8.5 ml/kg, the tidal volume the ventilator starts on, while ARDS and after RSI keep 6 to 8 ml/kg.

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## Paediatric scenarios

Child scenarios follow the Resuscitation Council UK (RCUK) Guidelines 2025: the Paediatric advanced life support and Paediatric cardiac arrhythmias algorithms, and the Paediatric emergency drug chart.

**WETFLAG.** The brief for a child scenario shows the WETFLAG values worked out from the child's weight. If you do not give a weight, it is estimated from age using the RCUK chart. WETFLAG starts hidden on the room monitor. To show it to the team, press **Show WETFLAG on Monitor** in the Rhythm & resus section; press **Hide WETFLAG on Monitor** to take it away again. To have it showing from the start, tick **Show WETFLAG on Monitor from the start** on the setup screen.

| Letter | Value used |
|---|---|
| W | Weight in kg |
| E | Defibrillation energy, 4 J/kg |
| T | Tracheal tube size (cuffed and uncuffed), from the RCUK chart |
| F | Fluid bolus 10 ml/kg, up to 500 ml |
| L | Lorazepam 0.1 mg/kg, up to 4 mg |
| A | Adrenaline 10 micrograms/kg, up to 1 mg |
| G | 10% glucose 2 ml/kg, up to 50 ml |

**Doses during the scenario.** When the team gives a drug to a child, the log shows the weight-based dose (for example atropine, adenosine, amiodarone, buccal midazolam, levetiracetam, 10% glucose and calcium gluconate). IM adrenaline for anaphylaxis follows the age bands. **Tools → Drug calculator** works out doses for any weight.

**Shocks for a child.** Defibrillation is 4 J/kg. From the 5th shock in refractory VF or pulseless VT, up to 8 J/kg is accepted without a flag. Cardioversion starts at 1 J/kg and doubles up to 4 J/kg.

**Newborns.** Newborn scenarios show the Newborn life support steps (Guidelines 2025) as prompts in the log.

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## The event log, notes and flags

Everything that happens is logged with its scenario time: treatments, obs changes, rhythm changes, shocks, monitoring on or off, and your notes. The log feeds the debrief.

- **Where:** the Event log section at the bottom right of the instructor screen, newest first. It stays on screen throughout. **Full log** (or **Tools → Full simulation log**) opens the whole log.
- **Add a note:** type in the box under the log and press **Add note**, for example "Team leader identified".
- **Add a flagged note:** type the note and press **Flag** instead.
- **Flag an existing entry:** click the small flag beside it. Click again to remove the flag.

Flagged entries appear in the **Flags and warnings** row of the debrief graph, so you can find the moments to discuss.

**Safety flags** count sequence deviations: actions done before their usual prerequisite was in place, such as an IV drug before IV access. Nothing is blocked; the team can still do it. When there is at least one, an amber **Safety flags** button appears in the top bar. It opens the list, with arrests, shocks and your flagged moments shown separately as significant events. Safety flags are not used in Quick Sim.

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## Finishing and the debrief

Press **Finish** (or `F`) and confirm. The debrief opens on your laptop only. The room monitor shows "Simulation complete" and asks the team to turn to you, with no score or diagnosis.

| Part of the debrief | What it shows |
|---|---|
| Performance Summary | Objectives met and the score (fully met, and with partial credit). The session's duration is at the top of the debrief. Quick Sim, Defib Sim and Ventilator Sim are not scored against objectives |
| Learning Objectives | Each objective as fully met, partly done or not in place, with the steps behind it |
| Obs, interventions and changes | The graph described below, at the top of the debrief |
| Sequence Deviations | The safety flags |

| Part of the debrief | What it shows |
|---|---|
| Defibrillation & Rhythm | Shown only when a shock was given: shocks, the shocks counted towards ROSC (a shock stacked within 5 s of the last one does not count), energies and each rhythm change |
| Defib Sim feedback | Defib Sim only: good practice and areas for improvement |
| Ventilator feedback | Any session that used the ventilator: when ventilation started, the pre-use check, settings against the scenario's targets, good practice, areas for improvement and each problem injected (see The ventilator and Ventilator Sim) |
| Simulation Log | Every entry with its time |
| Instructor Notes | A box for your own feedback notes |

Quick Sim has no scenario, so it has no score; the graph and the log still work.

## The obs graph

Every version (a full scenario, Quick Sim and Defib Sim) finishes with the same graph, **Obs, interventions and changes**. It shows the patient's true obs every 5 seconds over the whole session. What the team could see depended on what was attached.

- **Rhythm** along the top. Time without a pulse is red, and the same red shading runs down every chart below.
- One small chart each for heart rate, blood pressure (systolic and diastolic), SpO2 and respiratory rate. ETCO2 is added while capnography was attached. Temperature, glucose, pH and GCS are added only if they changed.
- Numbered markers underneath, in four rows: **Rhythm, arrest and shocks**; **Team actions and treatment**; **Facilitator changes** (the obs you changed or trended, with the values before and after, the patient's voice, presets, and the ventilator problems, lungs and settings you changed); and **Flags and warnings**. Markers that would overlap share one label, such as "5 +2".
- **What happened**, under the graph, lists every numbered marker with its time. Click one to mark it on the graph.
- Move the pointer over the graph, or click it and use the arrow keys, to read every value at that moment.
- **Show the obs as a table** lists the values every 30 seconds.

The graph records while the scenario clock runs. In Quick Sim it records from the moment the Quick Sim starts, without pressing START. Pressing START in Quick Sim still switches on the physiology clock, such as SpO2 falling when the patient is not breathing. In Defib Sim the clock starts at the learner's first action on the defib. In a full scenario, press START at the beginning or there will be no graph. The graph is kept if you reload the page and resume, and it is in the printed and downloaded report.

### Keeping a copy

- **Download Report** saves the debrief as a web page you can open on any computer, including the log and your instructor notes. Write your notes before you download.
- **Print** opens the report ready to print or save as PDF.
- **Certificate** (Defib Sim only) asks for the learner's name and prints a certificate of completion.

**Exit to Menu** returns to the setup screen. The session code and any linked screens stay the same, ready for the next scenario.

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## Tools, sound and shortcuts

The **Tools ▾** menu in the top bar holds the extras.

| Tool | What it does |
|---|---|
| Drug calculator (`D`) | Doses for any weight you enter; child doses follow the RCUK chart |
| Timer alerts (`T`) | Set a reminder at a scenario time, for example "5 mins: bloods back". It plays a tone, appears in the log, and is marked FIRED |
| Full simulation log | The whole event log in a window |
| Sound plays on | Choose where the pulse tone, alarms, the cuff noise, sound effects and the patient's voice play: room monitor (the default), this controller, or both. **Mute alarms** silences the pulse tone and alarms only. The defib tablet always makes its own sounds (charging, shock, metronome and, in Defib Sim, the cuff) |
| Mute / Unmute alarms | While muted, a red **Muted** button stays in the top bar so alarms are never silently off |
| Keyboard shortcuts (`?`) | The list below |

**Room monitor timer. Show Timer**, above the waveforms, puts a timer on the room monitor that you start and pause from your screen. It is useful for the team's own timings.

**Keyboard shortcuts** (they work straight after you click a button, except `Space`, which then presses that button again; they do nothing while you are typing in a box or a pop-up is open)

| Key | Action |
|---|---|
| `Space` | Pause or start the clock |

| Key | Action |
|---|---|
| `F` | Finish the scenario |
| `D` | Drug calculator |
| `T` | Timer alerts |
| `?` | Show the shortcuts |

## Troubleshooting

| Problem | Likely cause | Fix |
|---|---|---|
| Room monitor shows no numbers or "No sensor detected" | Monitoring not attached yet | Press **Attach**, or click the chips in Monitoring & access |
| A value you set is not on the room monitor | That sensor is not attached (your tile says "not on monitor") | Attach it when the team does |
| BP on the room monitor has not changed | The cuff has not cycled (this is deliberate) | Press **Cycle NIBP Now**, or use **Change and cycle cuff now** when you set the BP |
| No sound | Sound not enabled, muted, or playing on the other screen | Tap **Tap to Enable Sound** on the monitor; check for the red **Muted** button; check **Tools → Sound plays on**; check the device's volume |
| Badge says Sync error | The internet connection has dropped | Check the connection on each device |
| Room monitor or defib not following your changes | It is on a different session code, for example after **New code** | Join it again with the current code |
| Defib tablet asks for a code | It was opened without one | Type the session code, or scan the QR code from **Screens → Join by QR code** |
| Defib Sim log times all 00:00 | The clock has not started | Press **START**; it also starts at the learner's first press |
| Space bar does nothing, or presses a button | A button you clicked is still selected (Space presses it), or you are typing in a box or a pop-up is open | Click an empty part of the screen and try again; the other shortcuts work with a button selected |

| Problem | Likely cause | Fix |
|---|---|---|
| Instructor tab closed or crashed | — | Reopen the site and press **Resume** under "Resume Previous?" |
| The Restricted tab is locked | Restricted scenarios need an account with access | Sign in, then use **Request access** |
| A screen looks out of date after an update | The browser is showing the old version | Reload the page |